



System Test Specification
for the
KNEAD Example System
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Balance Project

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Distribution Statement D: Vinay Agarwal



DOCUMENT CHANGE HISTORY

The following table is a simple list of released revisions sent for review. Records of reviews and the review artifacts are saved with reviewer information in the The Balance Project artifact repository.

Change Record

Date	Version	Author(s)	Change Reference
13 Sep 2025	P1	Vinay Agarwal	Preliminary DRAFT version
5 Dec 2025	P1	Vinay Agarwal	Final Draft of the system

Draft P1 Preliminary version of this document.

1. Baseline Document draft 1
2. Final fix and update from the first draft



TABLE OF CONTENTS

DOCUMENT CHANGE HISTORY	i
TABLE OF CONTENTS	ii
LIST OF TABLES	v
LIST OF FIGURES	vi
CHAPTER	
1 Scope	1
1.1 Identification	1
1.2 System Overview	1
1.3 Document Overview	1
2 References	3
2.1 Acronyms and Abbreviations	3
2.2 Glossary and Definitions	3
2.3 Referenced Documents	3
2.3.1 External Documents	4
2.3.2 Project Specific Documents	4
3 Test Preparations	5
3.1	5
3.1.1 Hardware Preparation	5
3.1.2 Software Preparation	5
3.1.3 Other Preparation	5
3.2 Test2	5
3.2.1 Hardware Preparation	5
3.2.2 Software Preparation	5
3.2.3 Other Preparation	5
3.3 Test3	5
3.3.1 Hardware Preparation	6
3.3.2 Software Preparation	6
3.3.3 Other Preparation	6
4 Test Specifications	7
4.1 Test1	7
4.1.1 Case1	7
4.1.1.1 Requirements Addressed	7
4.1.1.2 Prerequisite Conditions	7
4.1.1.3 Inputs	7
4.1.1.4 Expected Outputs	7
4.1.1.5 Evaluation Criteria	7
4.1.1.6 Assumptions and Constraints	7
4.1.1.7 Procedure	7



4.1.2	Case2	11
4.1.2.1	Requirements Addressed	11
4.1.2.2	Prerequisite Conditions	11
4.1.2.3	Inputs	11
4.1.2.4	Expected Outputs	11
4.1.2.5	Evaluation Criteria	11
4.1.2.6	Assumptions and Constraints	12
4.1.2.7	Procedure	12
4.1.3	Case2	16
4.1.3.1	Requirements Addressed	16
4.1.3.2	Prerequisite Conditions	16
4.1.3.3	Inputs	16
4.1.3.4	Expected Outputs	16
4.1.3.5	Evaluation Criteria	16
4.1.3.6	Assumptions and Constraints	17
4.1.3.7	Procedure	17
4.2	Test2	21
4.2.1	Case1	21
4.2.1.1	Requirements Addressed	21
4.2.1.2	Prerequisite Conditions	21
4.2.1.3	Inputs	21
4.2.1.4	Expected Outputs	21
4.2.1.5	Evaluation Criteria	22
4.2.1.6	Assumptions and Constraints	22
4.2.1.7	Procedure	22
4.3	Test3	26
4.3.1	Case1	26
4.3.1.1	Requirements Addressed	26
4.3.1.2	Prerequisite Conditions	26
4.3.1.3	Inputs	26
4.3.1.4	Expected Outputs	26
4.3.1.5	Evaluation Criteria	27
4.3.1.6	Assumptions and Constraints	27
4.3.1.7	Procedure	27
4.3.2	Case2	31
4.3.2.1	Requirements Addressed	31
4.3.2.2	Prerequisite Conditions	31
4.3.2.3	Inputs	31
4.3.2.4	Expected Outputs	31
4.3.2.5	Evaluation Criteria	31
4.3.2.6	Assumptions and Constraints	32
4.3.2.7	Procedure	32
4.3.3	Case3	36
4.3.3.1	Requirements Addressed	36
4.3.3.2	Prerequisite Conditions	36



4.3.3.3	Inputs	36
4.3.3.4	Expected Outputs	36
4.3.3.5	Evaluation Criteria	36
4.3.3.6	Assumptions and Constraints	37
4.3.3.7	Procedure	37
5	Traceability	42
6	Notes	43
6.1	Note Area 1	43
6.2	Note Area 2	43
APPENDIX		
Additional Information		44
Index		45



LIST OF TABLES

Table		Page
1	Acronym Definitions	3
2	Glossary Terms and Definitions	3



LIST OF FIGURES

Figure		Page
1	System Overview	2



CHAPTER 1

Scope

This document provides the System Test Specification (**STS**) for the Balance System. The system will be referred to as the Balance System.

1.1 Identification

The Balance System described in this document shall be known as Balance System version 1. However, the **STS** described herein shall be applicable to pre-releases such as Beta-releases for a phased release as listed for each requirement.

1.2 System Overview

Figure 1 shows the high-level architecture for the Balance System system. This diagram shows the major external interfaces that provide the capabilities of Balance System.

The Balance System is a single player game. The game is a standalone game played on the STM32 board. This system would be a game where the user would have to balance a ball on a LCD screen that is builtin on the STM32 board. The objective of the game is to balance the ball on the screen based on the way the board was tilted. Using an onboard gyroscope that communicates across the board to the MCU though SPI communication. Balance System would keep track of the current position of the ball and where the next updated move is. This helps keep track of the system of where the ball is until a movement has occurred. Balance System shall process at a maximum 180 Hz. This would give the user enough time to process the current angle of the ball and be able to present on the LCD-TFT screen.

1.3 Document Overview

This section provides information about this document's security/privacy considerations, contents, structure, and version information.

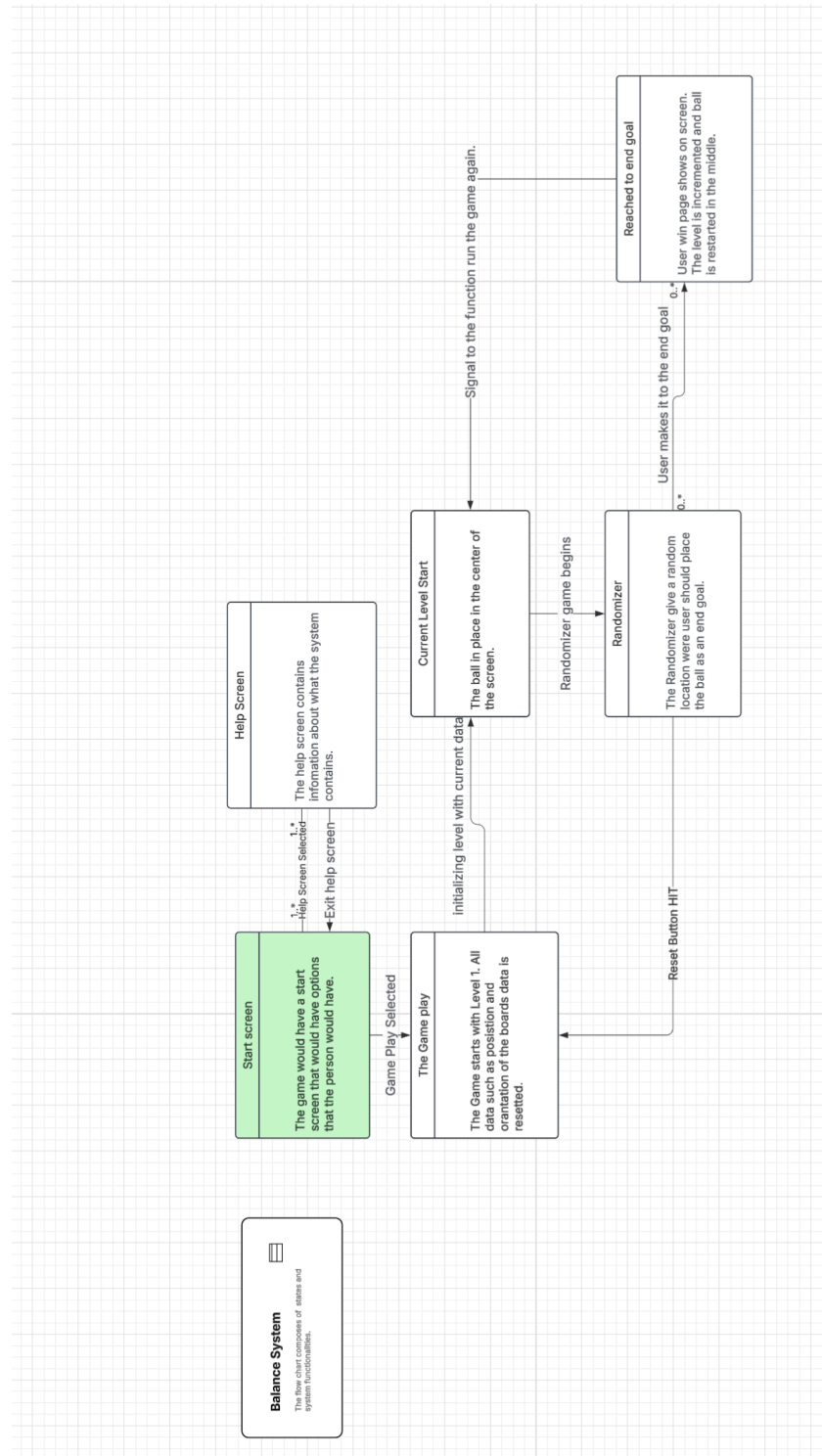


Figure 1: System Overview



CHAPTER 2

References

This section provides a list of referenced items for this document.

2.1 Acronyms and Abbreviations

This section defines acronyms and abbreviations used in this and related documents.

Table 1: Acronym Definitions

Acronym	Definition
YSM	Yourdon Structured Method
LTDC	Layered Transmission Display Controller
STM	STMicroelectronics
MEMS	Micro-Electro-Mechanical Systems
End of acronym definition table	

2.2 Glossary and Definitions

This section defines glossary terms used in this and related documents.

Table 2: Glossary Terms and Definitions

Glossary Term	Definition
STM32F429I	Micro-controller board has all component fit onto one board.
Customer	The professor that is view the grading all assignments.
End of glossary terms table	

2.3 Referenced Documents

This section lists the referenced documents for this document. The references are categorized into two categories:



External Documents not directly associated with this project.

Project Documents that are directly associated with this project.

2.3.1 External Documents

2.3.2 Project Specific Documents



CHAPTER 3

Test Preparations

This section describes the test preparations to be performed for each test.

3.1

Test is to check the board loads properly.

3.1.1 Hardware Preparation

This hardware preparation for test is to have the board the the micro-usb connected to a power source.

3.1.2 Software Preparation

This software preparation for test is to have the current version of the code loaded to the board.

3.1.3 Other Preparation

This other preparation for test is not necessary.

3.2 Test2

Test Test2 is to check if the help screen would show when pressed.

3.2.1 Hardware Preparation

This hardware preparation for test Test2 is to have the board the the micro-usb connected to a power source.

3.2.2 Software Preparation

This software preparation for test Test2 is to have the current version of the code loaded to the board.

3.2.3 Other Preparation

This other preparation for test Test2 is not nessary.

3.3 Test3

Test Test3 is to check if the game screen would show when pressed.



3.3.1 Hardware Preparation

This hardware preparation for test Test3 is to have the board the the micro-usb connected to a power source.

3.3.2 Software Preparation

This software preparation for test Test3 is to have the current version of the code loaded to the board.

3.3.3 Other Preparation

This other preparation for test Test3 is not necessary.



CHAPTER 4

Test Specifications

This section specifies the tests to be performed.

4.1 Test1

Test Test1 is .

4.1.1 Case1

Test Test1 case Case1 is to see if the system turns on.

4.1.1.1 Requirements Addressed

This requirements validated by test Test1 case Case1 are listed in § 4.1.1.7.

4.1.1.2 Prerequisite Conditions

This prerequisite conditions for test Test1 case Case1 are that there must need to be at least 5 volts to be applied to the system.

4.1.1.3 Inputs

This inputs for test Test1 case Case1 are listed in § 4.1.1.7.

4.1.1.4 Expected Outputs

This expected outputs for test Test1 case Case1 are listed in § 4.1.1.7.

4.1.1.5 Evaluation Criteria

This evaluation criteria for test Test1 case Case1 are listed in § 4.1.1.7.

4.1.1.6 Assumptions and Constraints

This procedure for test Test1 case Case1 is not necessary.

4.1.1.7 Procedure

This procedure for test Test1 case Case1 is **...TBD....**

See step 4.3.3.7.1 -6 for how to reference specific steps.

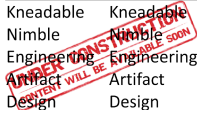
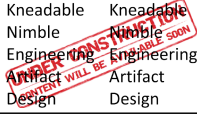


Test Procedure 4.1.1.7.1	
Test Procedure 1	
Requirements Validated by Test Procedure 4.1.1.7.1	
R-4.1.1.7.1 -1	Fake Rqmt 1
R-4.1.1.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.1.1.7.1	
N-4.1.1.7.1 -1	Note 1
N-4.1.1.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.1.1.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.1.1.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.1.1.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.1.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.1.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.1.1.7.1 – continues on the next page	



Test Procedure 4.1.1.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.1.1.7.1 – continues on the next page	



Test Procedure 4.1.1.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.1.1.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.1.1.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.1.1.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.1.1.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.1.1.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.1.1.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.1.1.7.1 – continues on the next page	



Test Procedure 4.1.1.7.1 – continued from previous page	
S-4.1.1.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.1.1.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.1.1.7.1	

4.1.2 Case2

Test Test1 case Case2 is the help screen showing up.

4.1.2.1 Requirements Addressed

This requirements validated by test Test1 case Case2 are listed in § 4.1.3.7.

4.1.2.2 Prerequisite Conditions

This prerequisite conditions for test Test1 case Case2 are not necessary.

4.1.2.3 Inputs

This inputs for test Test1 case Case2 are listed in § 4.1.3.7.

4.1.2.4 Expected Outputs

This expected outputs for test Test1 case Case2 are listed in § 4.1.3.7.

4.1.2.5 Evaluation Criteria

This evaluation criteria for test Test1 case Case2 are listed in § 4.1.3.7.



4.1.2.6 Assumptions and Constraints

This procedure for test Test1 case Case2 is not necessary.

4.1.2.7 Procedure

This procedure for test Test1 case Case2 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

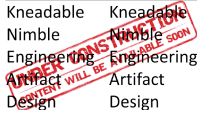
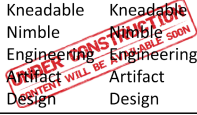


Test Procedure 4.1.2.7.1 Test Procedure 1	
Requirements Validated by Test Procedure 4.1.2.7.1	
R-4.1.2.7.1 -1	Fake Rqmt 1
R-4.1.2.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.1.2.7.1	
N-4.1.2.7.1 -1	Note 1
N-4.1.2.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.1.2.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.1.2.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.1.2.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.2.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.2.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.1.2.7.1 – continues on the next page	



Test Procedure 4.1.2.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.1.2.7.1 – continues on the next page	



Test Procedure 4.1.2.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.1.2.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.1.2.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.1.2.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.1.2.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.1.2.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.1.2.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.1.2.7.1 – continues on the next page	



Test Procedure 4.1.2.7.1 – continued from previous page	
S-4.1.2.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.1.2.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.1.2.7.1	

4.1.3 Case2

Test Test1 case Case2 is the game working.

4.1.3.1 Requirements Addressed

This requirements validated by test Test1 case Case2 are listed in § 4.1.3.7.

4.1.3.2 Prerequisite Conditions

This prerequisite conditions for test Test1 case Case2 are not necessary.

4.1.3.3 Inputs

This inputs for test Test1 case Case2 are listed in § 4.1.3.7.

4.1.3.4 Expected Outputs

This expected outputs for test Test1 case Case2 are listed in § 4.1.3.7.

4.1.3.5 Evaluation Criteria

This evaluation criteria for test Test1 case Case2 are listed in § 4.1.3.7.



4.1.3.6 Assumptions and Constraints

This procedure for test Test1 case Case2 is not necessary.

4.1.3.7 Procedure

This procedure for test Test1 case Case2 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

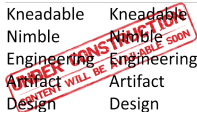
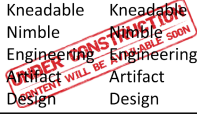


Test Procedure 4.1.3.7.1	
Test Procedure 1	
Requirements Validated by Test Procedure 4.1.3.7.1	
R-4.1.3.7.1 -1	Fake Rqmt 1
R-4.1.3.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.1.3.7.1	
N-4.1.3.7.1 -1	Note 1
N-4.1.3.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.1.3.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.1.3.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.1.3.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.3.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.3.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.1.3.7.1 – continues on the next page	



Test Procedure 4.1.3.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
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	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.1.3.7.1 – continues on the next page	



Test Procedure 4.1.3.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.1.3.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.1.3.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.1.3.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.1.3.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.1.3.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.1.3.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.1.3.7.1 – continues on the next page	



Test Procedure 4.1.3.7.1 – continued from previous page	
S-4.1.3.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.1.3.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.1.3.7.1	

4.2 Test2

Test Test2 is .

4.2.1 Case1

Test Test2 case Case1 is the help screen showing up.

4.2.1.1 Requirements Addressed

This requirements validated by test Test2 case Case1 are listed in § 4.2.1.7.

4.2.1.2 Prerequisite Conditions

This prerequisite conditions for test Test2 case Case1 are not necessary.

4.2.1.3 Inputs

This inputs for test Test2 case Case1 are listed in § 4.2.1.7.

4.2.1.4 Expected Outputs

This expected outputs for test Test2 case Case1 are listed in § 4.2.1.7.

**4.2.1.5 Evaluation Criteria**

This evaluation criteria for test Test2 case Case1 are listed in § 4.2.1.7.

4.2.1.6 Assumptions and Constraints

This procedure for test Test2 case Case1 is not necessary.

4.2.1.7 Procedure

This procedure for test Test2 case Case1 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

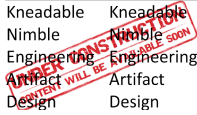
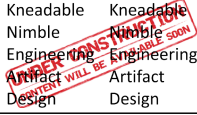


Test Procedure 4.2.1.7.1	
Test Procedure 1	
Requirements Validated by Test Procedure 4.2.1.7.1	
R-4.2.1.7.1 -1	Fake Rqmt 1
R-4.2.1.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.2.1.7.1	
N-4.2.1.7.1 -1	Note 1
N-4.2.1.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.2.1.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.2.1.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.2.1.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.2.1.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.2.1.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.2.1.7.1 – continues on the next page	



Test Procedure 4.2.1.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.2.1.7.1 – continues on the next page	



Test Procedure 4.2.1.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.2.1.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.2.1.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.2.1.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.2.1.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.2.1.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.2.1.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.2.1.7.1 – continues on the next page	



Test Procedure 4.2.1.7.1 – continued from previous page	
S-4.2.1.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.2.1.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.2.1.7.1	

4.3 Test3

Test Test3 is a system level test.

4.3.1 Case1

Test Test3 case Case1 is the game working.

4.3.1.1 Requirements Addressed

This requirements validated by test Test3 case Case1 are listed in § 4.3.1.7.

4.3.1.2 Prerequisite Conditions

This prerequisite conditions for test Test3 case Case1 are not necessary.

4.3.1.3 Inputs

This inputs for test Test3 case Case1 are listed in § 4.3.1.7.

4.3.1.4 Expected Outputs

This expected outputs for test Test3 case Case1 are listed in § 4.3.1.7.



4.3.1.5 Evaluation Criteria

This evaluation criteria for test Test3 case Case1 are listed in § 4.3.1.7.

4.3.1.6 Assumptions and Constraints

This procedure for test Test3 case Case1 is not necessary.

4.3.1.7 Procedure

This procedure for test Test3 case Case1 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

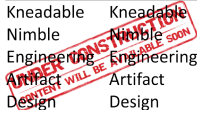
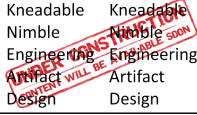


Test Procedure 4.3.1.7.1 Test Procedure 1	
Requirements Validated by Test Procedure 4.3.1.7.1	
R-4.3.1.7.1 -1	Fake Rqmt 1
R-4.3.1.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.3.1.7.1	
N-4.3.1.7.1 -1	Note 1
N-4.3.1.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.3.1.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.3.1.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.3.1.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.1.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.1.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.3.1.7.1 – continues on the next page	



Test Procedure 4.3.1.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.3.1.7.1 – continues on the next page	



Test Procedure 4.3.1.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.3.1.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.3.1.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.3.1.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.3.1.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.3.1.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.3.1.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.3.1.7.1 – continues on the next page	



Test Procedure 4.3.1.7.1 – continued from previous page	
S-4.3.1.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.3.1.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.3.1.7.1	

4.3.2 Case2

Test Test3 case Case2 is the help screen showing up.

4.3.2.1 Requirements Addressed

This requirements validated by test Test3 case Case2 are listed in § 4.3.2.7.

4.3.2.2 Prerequisite Conditions

This prerequisite conditions for test Test3 case Case2 are not necessary.

4.3.2.3 Inputs

This inputs for test Test3 case Case2 are listed in § 4.3.2.7.

4.3.2.4 Expected Outputs

This expected outputs for test Test3 case Case2 are listed in § 4.3.2.7.

4.3.2.5 Evaluation Criteria

This evaluation criteria for test Test3 case Case2 are listed in § 4.3.2.7.



4.3.2.6 Assumptions and Constraints

This procedure for test Test3 case Case2 is not necessary.

4.3.2.7 Procedure

This procedure for test Test3 case Case2 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

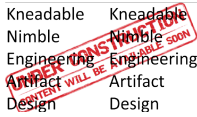
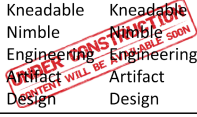


Test Procedure 4.3.2.7.1	
Test Procedure 1	
Requirements Validated by Test Procedure 4.3.2.7.1	
R-4.3.2.7.1 -1	Fake Rqmt 1
R-4.3.2.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.3.2.7.1	
N-4.3.2.7.1 -1	Note 1
N-4.3.2.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.3.2.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.3.2.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.3.2.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.2.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.2.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.3.2.7.1 – continues on the next page	



Test Procedure 4.3.2.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.3.2.7.1 – continues on the next page	



Test Procedure 4.3.2.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.3.2.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.3.2.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.3.2.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.3.2.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.3.2.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.3.2.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.3.2.7.1 – continues on the next page	



Test Procedure 4.3.2.7.1 – continued from previous page	
S-4.3.2.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.3.2.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.3.2.7.1	

4.3.3 Case3

Test Test3 case Case3 is the game working.

4.3.3.1 Requirements Addressed

This requirements validated by test Test3 case Case3 are listed in § 4.3.3.7.

4.3.3.2 Prerequisite Conditions

This prerequisite conditions for test Test3 case Case3 are not necessary.

4.3.3.3 Inputs

This inputs for test Test3 case Case3 are listed in § 4.3.3.7.

4.3.3.4 Expected Outputs

This expected outputs for test Test3 case Case3 are listed in § 4.3.3.7.

4.3.3.5 Evaluation Criteria

This evaluation criteria for test Test3 case Case3 are listed in § 4.3.3.7.



4.3.3.6 Assumptions and Constraints

This procedure for test Test3 case Case3 is not necessary.

4.3.3.7 Procedure

This procedure for test Test3 case Case3 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

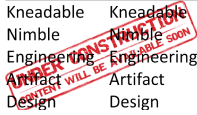
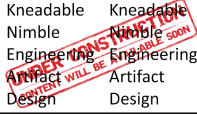


Test Procedure 4.3.3.7.1	
Test Procedure 1	
Requirements Validated by Test Procedure 4.3.3.7.1	
R-4.3.3.7.1 -1	Fake Rqmt 1
R-4.3.3.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.3.3.7.1	
N-4.3.3.7.1 -1	Note 1
N-4.3.3.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.3.3.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.3.3.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.3.3.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.3.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.3.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.3.3.7.1 – continues on the next page	



Test Procedure 4.3.3.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.3.3.7.1 – continues on the next page	



Test Procedure 4.3.3.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.3.3.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.3.3.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.3.3.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.3.3.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.3.3.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.3.3.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.3.3.7.1 – continues on the next page	



Test Procedure 4.3.3.7.1 – continued from previous page	
S-4.3.3.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.3.3.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.3.3.7.1	



CHAPTER 5

Traceability

ALL-TRACEABILITY :: THIS SECTION SHALL TRACEABILITY BETWEEN THE DEFINING REQUIREMENTS AND THE TESTS PLANNED TO PERFORM THE VALIDATION OF THE REQUIREMENTS. THIS ARTIFACT MAPS:

- (A) EACH TEST IDENTIFIED IN THIS ARTIFACT TO THE UNDERLYING REQUIREMENTS AND, IF APPLICABLE, SYSTEM REQUIREMENTS IT ADDRESSES. (ALTERNATIVELY, THIS TRACEABILITY MAY BE PROVIDED ELSEWHERE IN THIS ARTIFACT AND REFERENCED FROM THIS PARAGRAPH.), OR
- (B) FROM EACH CAPABILITY REQUIREMENT AND, IF APPLICABLE, EACH SUB-SYSTEM REQUIREMENT COVERED BY THIS ARTIFACT TO THE ARTIFACT(S) THAT ADDRESS IT. THE TRACEABILITY SHALL COVER THE REQUIREMENTS IN ALL-APPLICABLE ARTIFACTS OR, IF APPLICABLE, THE SYSTEM DESIGN REQUIREMENTS IN THE **SSS**.

This section provides traceability of the system components and interfaces to the design requirements. In general, the requirements are included in the individual tests, but they can be summarized here via automatic tools that read the test procedure files and generate this summary.



CHAPTER 6

Notes

ALL-NOTES :: THIS SECTION SHALL CONTAIN ANY GENERAL INFORMATION THAT AIDS IN UNDERSTANDING THIS ARTIFACT (E.G., BACKGROUND INFORMATION, RATIONALE, ETC.)

This chapter is explanations about how test are setup and how we use them. The most basic test is used to to if the system turns on.

6.1 Note Area 1

ALL-NOTES :: THIS SECTION SHALL CONTAIN ANY GENERAL INFORMATION THAT AIDS IN UNDERSTANDING THIS ARTIFACT (E.G., BACKGROUND INFORMATION, RATIONALE, ETC.)

This section is to see how to test and move to the help screen.

6.2 Note Area 2

ALL-NOTES :: THIS SECTION SHALL CONTAIN ANY GENERAL INFORMATION THAT AIDS IN UNDERSTANDING THIS ARTIFACT (E.G., BACKGROUND INFORMATION, RATIONALE, ETC.)

This section is see if the system can start the game and see if it works.



APPENDIX

Additional Information

ALL-APPENDIX :: THIS SECTION SHALL CONTAIN ANY GENERAL INFORMATION THAT AIDS IN UNDERSTANDING THIS ARTIFACT (E.G., BACKGROUND INFORMATION, RATIONALE, ETC.)

This section provides additional information, as necessary, to augment the **STS**.



Index

All To Be Determined Items, 7

Glossary
 Customer, 3

MIL-STD-498
 SSS, 42
 STD, 1, 44

This System, 1

Yourdon's Structured Method, 3